

ETHAN MAO

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EDUCATION

Bachelor of Applied Science in Electrical Engineering + PEY Co-op September 2024 - April 2029
University of Toronto

3.81 cGPA, Intended Minor: History

Relevant Courses: Digital Systems (Verilog), Computer Fundamentals (C), Programming Fundamentals (C++), Calculus III, Introductory Electronics, Advanced Engineering Mathematics, Circuit Analysis, Signals and Systems, Electric and Magnetic Fields 2, Computer Organization (Assembly), Hardware Design and Communication, Introduction to Control Systems, Fundamentals of Electrical Energy Systems, Analog Electronics, Introduction to Electronic Devices

WORK EXPERIENCE

Advanced Membranes Lab May 2026 - August 2026
Automation Engineering Research Student

- Conducted research of Self Driving Labs to automate the creation and testing of Reverse-Osmosis membranes
- Used CAD to model and 3D print parts to add functionalities such as humidity monitoring, cleaning, and additional solutions
- Improved reliability to 92%, allowing for experiments to run overnight without human oversight

Elections Poll Categorizer January 2025 - April 2025

- Canvassed 2000+ homes, directly engaging and speaking with hundreds of constituents
- Trained and organized volunteers, allowing for a larger campaign to reach more individuals
- Led teams of volunteers during large canvasses, ensuring the coordination, effectiveness, and safety of the group

PROJECTS

Hardware Design and Communication: Radio Filter, Mixer, and Amplifier Subsystem January 2026 - April 2026

- Designed a PCB in Altium to be modular and easily testable, allowing for quick diagnosing and resolving issues
- Designed, simulated, and prototyped a bandpass filter, ensuring the final design functioned without fail
- Implemented contingencies, allowing for the PCB to remain functional despite a faulty relay

Computer Organization: Blackjack Card Game March 2026

- Designed and tested a fully functional Blackjack game on the NiosV processor, written in C
- Wrote modular game logic functions, allowing for easy integration with I/O devices
- Implemented custom audio effects, elevating the game's experience with audio feedback and background music

Digital Systems: Virtual Piano November 2025

- Designed and developed a virtual piano using Verilog to be implemented on an FPGA board
- Developed the audio output to produce sound waves of variable frequencies, requiring no memory
- Created custom tests using ModelSim, debugging the program and ensuring proper functionality

Engineering Strategies and Practice 2: Packaging for Bio-Degradable Plant Pots January 2025 - April 2025

- Designed space efficient and economical package to store and ship plant pots produced by our Client
- Created a space-optimization program in C, maximizing the space-efficiency of the final proposed design
- Selected as one of 20 showcase teams for demonstrating exceptionality out of 187 total teams

Engineering Strategies and Practice: Courtyard Redesign

September 2024 - December 2024

- Effectively manage and organize workload and internal deadlines, ensuring fair distribution and on-time completion of work
- Facilitate team meetings, creating a professional and productive working environment

STUDENT GOVERNMENT AND EXTRA-CURRICULAR ACTIVITIES

University of Toronto Engineering Society

April 2025 - April 2026

Vice President Communications

- Managed email and social media communications between the Engineering Society and all 6,000+ engineering students
- Oversaw quality and content review for student yearbook, magazine, satirical newspaper, agenda, and first-year handbook, ensuring publication schedules
- Reduced printing costs by over \$5000 (30%) by negotiating with an alternative printing company for the student newspaper, allowing for additional publications of increased size
- Oversaw technical administrators to maintain the website and servers of the Engineering Society, ensuring that online resources are reliably available to students

University of Toronto Engineering Society

September 2024 - September 2025

First Year Chair

- Rescoped and defined the position to be responsible for community building and engagement, position was previously undefined and had little precedent due to previous years of inactivity
- Created and managed a team to build a wooden cart used in an annual chariot race, giving first-years an opportunity to learn woodworking skills while creating a sense of community
- Effectively managed a budget of \$1,000, funding initiatives and events for all first-year engineering students
- Advertised and hosted two first-year socials, boasting a total attendance of 150+ first-year students

Engineering Society Board of Directors

September 2024 - April 2025

First Year Representative

- Elected position, voted by all first-year engineering students
- Represent the needs and wants of all first-year students to Engineering Society Board of Directors, ensuring representation
- Converse with constituents to gauge opinions and gather ideas to effectively vote on policy

Electrical and Computer Engineering

September 2024 - Present

ECE Class Representative

- Elected position, voted by all electrical and computer engineering students in my year
- Advocate for class needs and issues to professors and faculty, creating a fairer learning environment
- Presented concerns to professors, allowing for assignment extensions when necessary

Engineering Society

September 2024 - April 2026

Policies and Structures Committee Member

- Create, draft, and revise bylaws in the Engineering Society Constitution
- Convene monthly to debate and discuss policy changes to uphold the ethics and integrity of the Engineering Society

University of Toronto Flight Endurance Team

May 2025 - Present

Operations Manager

- Founding member of the University of Toronto Flight Endurance Team, a design team which aims to design and create Unmanned Aerial Vehicles capable of long-term flights

- Managed a team of 30+ members, ensuring proper and effective communication and collaboration between internal divisions
- Oversaw technical divisions, ensuring progress of the team and effective work allocation

ChemE Cube

January 2026 - Present

PCB Design Lead

- Designed and assembled a PCB to connect different I/O devices, controlling the functionality of the carbon-capture cube
- Created a modular design with cold-swappable components
- Collaborated with mechanical and process teams to ensure design met functionality requirements and physical constraints

Iron Pin Ceremony

August 2025 - April 2026

Outreach Director

- Organized tabling events, raising awareness regarding the new ceremony
- Created guest speaker lists, allowing students to hear from Industry Professionals about engineering ethics

University of Toronto Engineering Competitions

May 2025 - January 2026

Parliamentary Debate Director

- Organized the Parliamentary Debate Competition, one of the seven categories in the annual engineering competition
- Engaged in marketing, outreach, and workshop events, doubling Debate attendance from the previous record
- Created and tested unique motions related to the theme of healthcare accessibility, ensuring debaters had relevant, nuanced, and balanced motions

TULA B

September 2025 - November 2025

Team Member

- Trained as a team twice a week, practicing Ultimate Frisbee
- Compete monthly against other universities

University of Toronto Aerospace Team Rocketry

September 2024 - April 2025

Avionics Division General Member

- Design PCBs using Altium for the UTAT solid, hybrid, and liquid rockets
- Effectively turn high-level architectural drawings into PCB Schematics
- Created Power and Battery Management Test Systems, enabling safe operation of the rockets

Human Powered Vehicles Design Team

September 2024 - April 2025

General Member

- Perform Carbon Fiber layups for repairs and building vehicles
- Create prototypes to test concepts for construction

Hide and Seek Club

September 2024 - April 2026

Executive Member

- Organize games for 100+ members, providing a fun environment to de-stress
- Wrote the club's policy document, outlining club rules and protocols for emergency scenarios

Intramural Ultimate Frisbee

September 2024 - Present

Team Member

- Compete as a team in weekly games against other intramural teams

SKILLS

Programming: C/C++, Verilog, RISC-V Assembly, Python

CAD: Fusion 360, Altium Designer, KiCad

Soft Skills: Leadership, Event Planning, Organization, Communication, Time Management, Public Speaking and Relations, Advertising and Promotion, Logistics, Team Management

Technical Skills: Microsoft Teams, Carbon Fiber Layups, Creating and Managing Budgets, Formal Writing, ModelSim

AWARDS AND FELLOWSHIPS

Enbridge Scholarship in Engineering June 2026
\$5,700 given to a single second year student who demonstrates significant community involvement and volunteer work

Dean's Undergraduate Student Summer Research Pivot Fellowship March 2026
\$8,000 given to 16 undergraduate students, recognizing academic merit and intended to support co-supervised research

Engineering Class of 5T6 Award Of Merit June 2025
\$24,000 awarded to a single first year engineering student for exceptional academics & community involvement

Class of 5T0 Engineering Leadership Award June 2025
\$10,000 granted to a single incoming second year engineering student based on academic achievement and leadership